
Linear Algebra Foundations

STAT 9120: Theory of Linear Models

Fall 2026

Purpose. These notes provide the linear algebra needed to begin a rigorous study of linear model theory. They are written to be accessible to students who have not taken a formal linear algebra course. The emphasis is on the concepts that recur throughout the course: column spaces, null spaces, rank, orthogonality, projections, eigenvalues, and quadratic forms.

The guiding statistical model is

$$Y = X\beta + \varepsilon.$$

Contents

1	Why Linear Algebra Matters for Linear Models	3
2	Vectors and Linear Combinations	4
3	Matrices and Matrix Multiplication	4
3.1	Matrix-vector multiplication	5
3.2	Matrix multiplication	5
4	Transpose and Symmetry	6
5	Systems of Linear Equations	6
6	Span and Subspaces	7
7	The Column Space	7
8	Linear Independence	7
9	Basis, Dimension, and Rank	8
9.1	Full column rank	9
10	The Null Space and Rank-Nullity	9
11	Identity Matrices and Inverses	9
12	A Fundamental Result for $X^T X$	10
13	Inner Products, Norms, and Orthogonality	11
14	Orthogonal Complements	11
15	Orthogonal Projection	12
15.1	Projection onto a single vector	12
16	Least Squares as Projection	12
17	Projection Matrices	13
18	The Residual Projection	14
19	Eigenvalues and Eigenvectors	14

20 Symmetric Matrices and the Spectral Theorem	15
21 Eigenvalues of Projection Matrices	15
22 Trace and Rank of a Projection	16
23 Quadratic Forms	16
24 Positive Semidefinite and Positive Definite Matrices	17
24.1 The matrix $X^T X$	17
25 Quadratic Forms and Eigenvalues	17
26 Generalized Inverses: A Preview	18
27 Worked Regression Example	18
28 Worked Rank-Deficient Example	19
29 Summary of Core Relationships	19
30 Results Worth Being Able to Prove	20
31 Practice Problems	20
32 Looking Ahead	22

1. Why Linear Algebra Matters for Linear Models

The general linear model is written as

$$Y = X\beta + \varepsilon,$$

where

- $Y \in \mathbb{R}^n$ is the random response vector,
- $X \in \mathbb{R}^{n \times p}$ is the design matrix,
- $\beta \in \mathbb{R}^p$ is a vector of unknown parameters, and
- $\varepsilon \in \mathbb{R}^n$ is a random error vector.

Nearly every theoretical question about this model has a linear algebra component. For example:

- When does $X\beta = \mu$ have a unique solution?
- What does it mean for the columns of X to be linearly dependent?
- Why is least squares an orthogonal projection?
- Why does $X(X^T X)^{-1} X^T$ appear in the full-rank model?
- What changes when $X^T X$ is not invertible?
- Why do quadratic forms such as $Y^T A Y$ appear in sums of squares and hypothesis tests?

Key idea

The most useful viewpoint in this course is not that a matrix is merely a rectangular array of numbers. A matrix represents a *linear transformation*, and its columns determine a geometric subspace. The theory of linear models is largely the study of how random vectors interact with these subspaces.

Learning objectives

By the end of these notes, you should be able to:

1. interpret matrix-vector multiplication as a linear combination of columns;
2. describe column spaces, null spaces, bases, dimensions, and rank;
3. recognize the relationship between rank and identifiability in a linear model;
4. use inner products and orthogonality to describe least squares geometrically;
5. identify and work with orthogonal projection matrices;
6. explain the role of eigenvalues and the spectral theorem for symmetric matrices;
7. work with quadratic forms and positive (semi)definite matrices; and
8. understand why generalized inverses are needed for rank-deficient models.

2. Vectors and Linear Combinations

Definition 2.1 (Vector). A vector in $\mathbb{R}^n$ is an ordered collection of n real numbers, usually written as a column:

$$x = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}.$$

For example,

$$x = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} \in \mathbb{R}^3.$$

The basic operations are vector addition and scalar multiplication. If $x, y \in \mathbb{R}^n$ and $c \in \mathbb{R}$, then

$$x + y = \begin{pmatrix} x_1 + y_1 \\ \vdots \\ x_n + y_n \end{pmatrix}, \quad cx = \begin{pmatrix} cx_1 \\ \vdots \\ cx_n \end{pmatrix}.$$

Definition 2.2 (Linear combination). A linear combination of vectors $x_1, \dots, x_p$ is any vector of the form

$$c_1x_1 + \dots + c_px_p,$$

where $c_1, \dots, c_p \in \mathbb{R}$.

Example 2.3. Let

$$x_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \quad x_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}.$$

Then

$$2x_1 - 3x_2 = \begin{pmatrix} 2 \\ -3 \\ -1 \end{pmatrix}.$$

Linear model connection

The mean vector of the linear model is $E(Y) = X\beta$. Once we view X by its columns, $X\beta$ is simply a linear combination of those columns. This observation motivates the column-space representation of a linear model.

3. Matrices and Matrix Multiplication

An $n \times p$ matrix is an array with n rows and p columns:

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1p} \\ a_{21} & a_{22} & \cdots & a_{2p} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{np} \end{pmatrix} \in \mathbb{R}^{n \times p}.$$

It is often more useful to regard A as a collection of columns,

$$A = (a_1 \ a_2 \ \cdots \ a_p), \quad a_j \in \mathbb{R}^n.$$

3.1 Matrix-vector multiplication

If $A \in \mathbb{R}^{n \times p}$ and $x \in \mathbb{R}^p$, then $Ax \in \mathbb{R}^n$. Writing

$$x = \begin{pmatrix} x_1 \\ \vdots \\ x_p \end{pmatrix},$$

we have

$$Ax = x_1 a_1 + \cdots + x_p a_p.$$

This identity is one of the most important in the course: multiplying a matrix by a vector forms a linear combination of the matrix columns.

Example 3.1. Let

$$A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{pmatrix}, \quad x = \begin{pmatrix} 3 \\ 2 \end{pmatrix}.$$

Then

$$Ax = 3 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + 2 \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 3 \\ 5 \\ 7 \end{pmatrix}.$$

For simple linear regression,

$$X = \begin{pmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{pmatrix}, \quad \beta = \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix}.$$

Hence

$$X\beta = \beta_0 \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix} + \beta_1 \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}.$$

3.2 Matrix multiplication

If $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times p}$, then

$$AB \in \mathbb{R}^{m \times p}.$$

A useful dimension check is

$$(m \times n)(n \times p) = (m \times p).$$

Matrix multiplication is generally *not commutative*:

$$AB \neq BA.$$

One product may even be defined when the other is not. Matrix multiplication is, however, associative:

$$A(BC) = (AB)C.$$

4. Transpose and Symmetry

Definition 4.1 (Transpose). For a matrix A , the transpose A^T is obtained by interchanging rows and columns.

For example,

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \implies A^T = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix}.$$

Important identities include

$$(A^T)^T = A, \quad (A + B)^T = A^T + B^T, \quad (AB)^T = B^T A^T.$$

Notice the reversal of order in the last identity.

Definition 4.2 (Symmetric matrix). A square matrix A is symmetric if

$$A^T = A.$$

Two matrices that appear constantly in linear model theory are $X^T X$ and XX^T . Both are symmetric because, for example,

$$(X^T X)^T = X^T (X^T)^T = X^T X.$$

5. Systems of Linear Equations

A system of linear equations can be written compactly as

$$Ax = b.$$

For example,

$$\begin{aligned} x_1 + x_2 &= 3, \\ 2x_1 - x_2 &= 0 \end{aligned}$$

is equivalent to

$$\begin{pmatrix} 1 & 1 \\ 2 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 3 \\ 0 \end{pmatrix}.$$

A linear system can have a unique solution, infinitely many solutions, or no solution.

Because Ax is a linear combination of the columns of A , the equation $Ax = b$ asks a geometric question:

Can b be written as a linear combination of the columns of A ?

This leads to the concepts of span and column space.

6. Span and Subspaces

Definition 6.1 (Span). Given vectors $x_1, \dots, x_p$, their span is

$$\text{span}(x_1, \dots, x_p) = \{c_1x_1 + \dots + c_px_p : c_1, \dots, c_p \in \mathbb{R}\}.$$

Definition 6.2 (Subspace). A subset $S \subseteq \mathbb{R}^n$ is a linear subspace if

1. $0 \in S$;
2. if $x, y \in S$, then $x + y \in S$;
3. if $x \in S$ and $c \in \mathbb{R}$, then $cx \in S$.

Examples include $\{0\}$, any line through the origin, any plane through the origin, and all of $\mathbb{R}^n$.

A line or plane that does *not* pass through the origin is not a subspace.

7. The Column Space

Definition 7.1 (Column space). For

$$A = \begin{pmatrix} a_1 & \cdots & a_p \end{pmatrix},$$

the column space of A is

$$\mathcal{C}(A) = \text{span}(a_1, \dots, a_p).$$

Equivalently,

$$\boxed{\mathcal{C}(A) = \{Ax : x \in \mathbb{R}^p\}}.$$

The equation $Ax = b$ has at least one solution precisely when $b \in \mathcal{C}(A)$.

Linear model connection

For $Y = X\beta + \varepsilon$,

$$E(Y) = X\beta \in \mathcal{C}(X).$$

As β varies over $\mathbb{R}^p$, the set of all possible mean vectors is exactly $\mathcal{C}(X)$. Thus a linear model can be viewed geometrically as the statement

$$E(Y) \in \mathcal{C}(X).$$

The subspace $\mathcal{C}(X)$ is often called the *model space*.

8. Linear Independence

Definition 8.1 (Linear independence). Vectors $x_1, \dots, x_p$ are linearly independent if

$$c_1x_1 + \dots + c_px_p = 0$$

implies

$$c_1 = \dots = c_p = 0.$$

If a nonzero collection of coefficients satisfies the equation, the vectors are linearly dependent.

Example 8.2. Consider

$$x_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad x_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad x_3 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Since $x_3 = x_1 + x_2$, we have

$$x_1 + x_2 - x_3 = 0,$$

so the three vectors are linearly dependent.

Linear model connection

If the columns of X are linearly dependent, there exists $a \neq 0$ such that $Xa = 0$. Consequently,

$$X(\beta + a) = X\beta + Xa = X\beta.$$

Two different parameter vectors therefore generate the same mean vector. This is the basic source of nonidentifiability in rank-deficient linear models.

9. Basis, Dimension, and Rank

Definition 9.1 (Basis). A basis for a subspace S is a collection of vectors that both spans S and is linearly independent.

Definition 9.2 (Dimension). The dimension of a subspace S , denoted $\dim(S)$, is the number of vectors in any basis for S .

For example, the standard basis vectors

$$e_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad e_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad e_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

form a basis of $\mathbb{R}^3$, so $\dim(\mathbb{R}^3) = 3$.

Definition 9.3 (Rank). The rank of a matrix A is the dimension of its column space:

$$\boxed{\text{rank}(A) = \dim \mathcal{C}(A).}$$

Equivalently, the rank is the maximum number of linearly independent columns of A . A fundamental theorem states that

$$\text{rank}(A) = \text{rank}(A^T),$$

so the maximum number of linearly independent columns also equals the maximum number of linearly independent rows.

If $A \in \mathbb{R}^{n \times p}$, then

$$\text{rank}(A) \leq \min(n, p).$$

9.1 Full column rank

An $n \times p$ matrix A has full column rank if

$$\text{rank}(A) = p.$$

Equivalently, its p columns are linearly independent.

For a design matrix $X \in \mathbb{R}^{n \times p}$, full column rank is the usual full-rank condition in linear model theory.

10. The Null Space and Rank-Nullity

Definition 10.1 (Null space). The null space of $A \in \mathbb{R}^{n \times p}$ is

$$\mathcal{N}(A) = \{x \in \mathbb{R}^p : Ax = 0\}.$$

Example 10.2. Let

$$A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}.$$

Then $Ax = 0$ is equivalent to

$$x_1 + 2x_2 = 0.$$

Thus

$$x = t \begin{pmatrix} -2 \\ 1 \end{pmatrix}, \quad t \in \mathbb{R},$$

and

$$\mathcal{N}(A) = \text{span} \left\{ \begin{pmatrix} -2 \\ 1 \end{pmatrix} \right\}.$$

Theorem 10.3 (Rank-nullity). If $A \in \mathbb{R}^{n \times p}$, then

$$\text{rank}(A) + \dim \mathcal{N}(A) = p.$$

The theorem separates the p directions in the domain into directions that contribute to Ax and directions that are annihilated by A .

Linear model connection

For a design matrix X , a vector $a \in \mathcal{N}(X)$ represents a parameter direction that does not change the mean:

$$X(\beta + a) = X\beta.$$

Thus $\dim \mathcal{N}(X)$ counts the number of linearly independent parameter directions that the mean structure cannot identify.

11. Identity Matrices and Inverses

The $n \times n$ identity matrix is

$$I_n = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}.$$

It satisfies

$$I_n x = x, \quad AI = A, \quad IA = A$$

whenever the dimensions are compatible.

Definition 11.1 (Inverse). For a square matrix A , an inverse A^{-1} is a matrix satisfying

$$AA^{-1} = A^{-1}A = I.$$

Not every square matrix is invertible. For an $n \times n$ matrix A , the following statements are equivalent:

1. A is invertible;
2. $\text{rank}(A) = n$;
3. $\mathcal{N}(A) = \{0\}$;
4. $Ax = b$ has a unique solution for every $b \in \mathbb{R}^n$.

Key idea

In this course, the conceptual question “is the matrix invertible?” is usually more important than computing an inverse by hand. We will rarely use determinant or cofactor formulas for inverses.

12. A Fundamental Result for $X^T X$

Proposition 12.1. For any real matrix X ,

$$\mathcal{N}(X^T X) = \mathcal{N}(X).$$

Proof. If $a \in \mathcal{N}(X)$, then $Xa = 0$, so immediately

$$X^T Xa = X^T 0 = 0.$$

Thus $\mathcal{N}(X) \subseteq \mathcal{N}(X^T X)$.

Conversely, suppose $a \in \mathcal{N}(X^T X)$. Then

$$X^T Xa = 0.$$

Premultiplying by a^T gives

$$a^T X^T Xa = 0.$$

But

$$a^T X^T Xa = (Xa)^T (Xa) = \|Xa\|^2.$$

Therefore $\|Xa\|^2 = 0$, which implies $Xa = 0$. Hence $a \in \mathcal{N}(X)$. □

Proposition 12.2. For any real matrix X ,

$$\text{rank}(X^T X) = \text{rank}(X).$$

Proof. If X has p columns, rank-nullity gives

$$\text{rank}(X^T X) = p - \dim \mathcal{N}(X^T X) = p - \dim \mathcal{N}(X) = \text{rank}(X).$$

□

It follows immediately that for $X \in \mathbb{R}^{n \times p}$,

$$\text{rank}(X) = p \iff X^T X \text{ is invertible.}$$

13. Inner Products, Norms, and Orthogonality

Definition 13.1 (Inner product). For $x, y \in \mathbb{R}^n$, the standard inner product is

$$\langle x, y \rangle = x^T y = \sum_{i=1}^n x_i y_i.$$

Definition 13.2 (Euclidean norm). The Euclidean norm of x is

$$\|x\| = \sqrt{x^T x}.$$

Thus $\|x\|^2 = x^T x$.

The least-squares objective can therefore be written

$$\|Y - X\beta\|^2 = (Y - X\beta)^T (Y - X\beta).$$

Definition 13.3 (Orthogonality). Vectors x and y are orthogonal, written $x \perp y$, if

$$x^T y = 0.$$

For example,

$$x = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad y = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

are orthogonal because $x^T y = 1 - 1 = 0$.

14. Orthogonal Complements

Definition 14.1 (Orthogonal complement). For a subspace $S \subseteq \mathbb{R}^n$, its orthogonal complement is

$$S^\perp = \{y \in \mathbb{R}^n : y^T x = 0 \text{ for every } x \in S\}.$$

Every $y \in \mathbb{R}^n$ can be uniquely decomposed as

$$y = y_S + y_{S^\perp},$$

where $y_S \in S$ and $y_{S^\perp} \in S^\perp$. Symbolically,

$$\boxed{\mathbb{R}^n = S \oplus S^\perp.}$$

Proposition 14.2. For any matrix X ,

$$\boxed{\mathcal{C}(X)^\perp = \mathcal{N}(X^T).}$$

Proof. A vector y belongs to $\mathcal{C}(X)^\perp$ precisely when it is orthogonal to every column of X . This is equivalent to saying that every entry of $X^T y$ is zero, i.e.,

$$X^T y = 0.$$

Hence $y \in \mathcal{N}(X^T)$. □

Linear model connection

This identity explains the normal equations. Least-squares residuals lie in $\mathcal{C}(X)^\perp$, so they satisfy

$$X^T e = 0.$$

15. Orthogonal Projection

Suppose S is a subspace of $\mathbb{R}^n$. Every $y \in \mathbb{R}^n$ has a unique decomposition

$$y = \hat{y} + e,$$

where

$$\hat{y} \in S, \quad e \in S^\perp.$$

Definition 15.1 (Orthogonal projection). The vector $\hat{y}$ is the orthogonal projection of y onto S , denoted

$$\hat{y} = P_S y.$$

Geometrically, $P_S y$ is the point in S closest to y :

$$P_S y = \arg \min_{z \in S} \|y - z\|^2.$$

15.1 Projection onto a single vector

If $u \neq 0$, the projection of y onto $\text{span}(u)$ is

$$\text{proj}_u(y) = \frac{u^T y}{u^T u} u.$$

If u is a unit vector, so $u^T u = 1$, this simplifies to

$$\text{proj}_u(y) = (u^T y) u.$$

This one-dimensional formula is the prototype for the general projection formulas used in regression.

16. Least Squares as Projection

Consider

$$Y = X\beta + \varepsilon.$$

Least squares minimizes

$$\|Y - X\beta\|^2.$$

Since $X\beta \in \mathcal{C}(X)$, this is exactly the problem of finding the point in $\mathcal{C}(X)$ closest to Y . Therefore

$$\hat{Y} = P_{\mathcal{C}(X)} Y.$$

The residual is

$$e = Y - \hat{Y}.$$

Because $e \in \mathcal{C}(X)^\perp$,

$$X^T e = 0.$$

Substituting $e = Y - X\hat{\beta}$ gives the *normal equations*:

$$X^T X \hat{\beta} = X^T Y.$$

If X has full column rank, $X^T X$ is invertible, and hence

$$\hat{\beta} = (X^T X)^{-1} X^T Y.$$

The fitted values are

$$\hat{Y} = X\hat{\beta} = X(X^T X)^{-1} X^T Y.$$

Define

$$H = X(X^T X)^{-1} X^T.$$

Then

$$\hat{Y} = HY.$$

17. Projection Matrices

Definition 17.1 (Idempotent matrix). A square matrix P is idempotent if

$$P^2 = P.$$

An orthogonal projection matrix is characterized by two properties:

$$P^T = P, \quad P^2 = P.$$

That is, an orthogonal projection matrix is symmetric and idempotent.

For

$$H = X(X^T X)^{-1} X^T,$$

we have

$$H^T = H,$$

and

$$\begin{aligned} H^2 &= X(X^T X)^{-1} X^T X(X^T X)^{-1} X^T \\ &= X(X^T X)^{-1} (X^T X) (X^T X)^{-1} X^T \\ &= X(X^T X)^{-1} X^T \\ &= H. \end{aligned}$$

Therefore H is the orthogonal projection matrix onto $\mathcal{C}(X)$.

Proposition 17.2. If $P = P^T = P^2$, then P is the orthogonal projection onto $\mathcal{C}(P)$.

Proof. For any y , write

$$y = Py + (I - P)y.$$

Clearly $Py \in \mathcal{C}(P)$. Also,

$$P(I - P)y = (P - P^2)y = 0,$$

so $(I - P)y \in \mathcal{N}(P)$. For symmetric P ,

$$\mathcal{N}(P) = \mathcal{C}(P)^\perp.$$

Hence the decomposition is orthogonal, and Py is the orthogonal projection of y onto $\mathcal{C}(P)$. $\square$

18. The Residual Projection

If P projects onto a subspace S , then $I - P$ projects onto $S^\perp$. Indeed,

$$(I - P)^2 = I - 2P + P^2 = I - P,$$

and if P is symmetric, so is $I - P$.

For the full-rank linear model, define

$$M = I - H.$$

Then M projects onto $\mathcal{C}(X)^\perp$, and

$$e = MY = (I - H)Y.$$

Thus

$$Y = HY + (I - H)Y = \hat{Y} + e,$$

with

$$\hat{Y} \perp e.$$

By the Pythagorean theorem,

$$\|Y\|^2 = \|\hat{Y}\|^2 + \|e\|^2,$$

or equivalently,

$$Y^T Y = Y^T H Y + Y^T (I - H) Y.$$

This geometric decomposition is the prototype for many analysis-of-variance decompositions.

19. Eigenvalues and Eigenvectors

Definition 19.1 (Eigenvalue and eigenvector). Let A be an $n \times n$ matrix. A nonzero vector v is an eigenvector of A if

$$Av = \lambda v$$

for some scalar λ . The scalar λ is the corresponding eigenvalue.

An eigenvector identifies a direction that is not rotated away from itself by A ; the matrix only stretches, shrinks, or reverses that direction.

Example 19.2. Let

$$A = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}.$$

Then

$$A \begin{pmatrix} 1 \\ 0 \end{pmatrix} = 2 \begin{pmatrix} 1 \\ 0 \end{pmatrix},$$

so $(1, 0)^T$ is an eigenvector with eigenvalue 2. Similarly, $(0, 1)^T$ has eigenvalue 3.

20. Symmetric Matrices and the Spectral Theorem

Symmetric matrices are especially important because their eigenstructure is well behaved.

Theorem 20.1 (Spectral theorem). *If $A = A^T$ is an $n \times n$ real symmetric matrix, then*

1. *all eigenvalues of A are real;*
2. *eigenvectors corresponding to distinct eigenvalues are orthogonal; and*
3. *there exists an orthogonal matrix Q and diagonal matrix Λ such that*

$$A = Q\Lambda Q^T.$$

An orthogonal matrix Q satisfies

$$Q^T Q = Q Q^T = I,$$

so

$$Q^{-1} = Q^T.$$

The diagonal entries of Λ are the eigenvalues of A , and the columns of Q are corresponding orthonormal eigenvectors.

Linear model connection

Projection matrices, covariance matrices, $X^T X$, and many matrices defining sums of squares are symmetric. The spectral theorem therefore provides one of the main tools for studying quadratic forms and their distributions.

21. Eigenvalues of Projection Matrices

Proposition 21.1. *If $P^2 = P$, every eigenvalue of P is either 0 or 1.*

Proof. Suppose $Pv = \lambda v$ for $v \neq 0$. Then

$$P^2 v = P(\lambda v) = \lambda P v = \lambda^2 v.$$

But $P^2 = P$, so also

$$P^2 v = P v = \lambda v.$$

Therefore

$$\lambda^2 = \lambda,$$

which implies

$$\lambda(\lambda - 1) = 0.$$

Hence $\lambda \in \{0, 1\}$. □

For an orthogonal projection, the eigenspace corresponding to eigenvalue 1 is the subspace onto which we project, while the eigenspace corresponding to eigenvalue 0 is its orthogonal complement.

22. Trace and Rank of a Projection

Definition 22.1 (Trace). For an $n \times n$ matrix A , the trace is

$$\text{tr}(A) = \sum_{i=1}^n a_{ii}.$$

A fundamental fact is that the trace equals the sum of the eigenvalues, counting multiplicities:

$$\text{tr}(A) = \sum_{i=1}^n \lambda_i.$$

If P is idempotent, all its eigenvalues are 0 or 1. The rank equals the number of nonzero eigenvalues, so

$$\boxed{\text{tr}(P) = \text{rank}(P)}.$$

If X has full column rank p , the hat matrix H projects onto the p -dimensional space $\mathcal{C}(X)$, and hence

$$\text{rank}(H) = \text{tr}(H) = p.$$

Similarly,

$$\text{rank}(I - H) = \text{tr}(I - H) = n - p.$$

These dimensions become regression and residual degrees of freedom.

23. Quadratic Forms

Definition 23.1 (Quadratic form). For $x \in \mathbb{R}^n$ and an $n \times n$ matrix A , an expression of the form

$$x^T A x$$

is called a quadratic form.

For example, if

$$A = \begin{pmatrix} a & b \\ b & c \end{pmatrix}, \quad x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix},$$

then

$$x^T A x = ax_1^2 + 2bx_1x_2 + cx_2^2.$$

Quadratic forms occur throughout linear model theory. For example,

$$\|Y - X\beta\|^2 = (Y - X\beta)^T(Y - X\beta),$$

and the residual sum of squares is

$$SSE = Y^T(I - H)Y.$$

Key idea

For any square matrix A ,

$$x^T A x = x^T \left(\frac{A + A^T}{2} \right) x.$$

Thus every real quadratic form can be represented by a symmetric matrix. This is one reason symmetric matrices dominate the theory of quadratic forms.

24. Positive Semidefinite and Positive Definite Matrices

Definition 24.1 (Positive semidefinite). A symmetric matrix A is positive semidefinite if

$$x^T A x \geq 0$$

for every x . We write $A \succeq 0$.

Definition 24.2 (Positive definite). A symmetric matrix A is positive definite if

$$x^T A x > 0$$

for every nonzero x . We write $A \succ 0$.

24.1 The matrix $X^T X$

For any $a \in \mathbb{R}^p$,

$$a^T X^T X a = (Xa)^T (Xa) = \|Xa\|^2 \geq 0.$$

Therefore

$$\boxed{X^T X \succeq 0.}$$

Moreover, $X^T X$ is positive definite exactly when $Xa = 0$ implies $a = 0$, i.e., exactly when X has full column rank:

$$\boxed{X^T X \succ 0 \iff \text{rank}(X) = p.}$$

25. Quadratic Forms and Eigenvalues

Suppose $A = A^T$ and use its spectral decomposition

$$A = Q \Lambda Q^T.$$

Set

$$z = Q^T x.$$

Then

$$\begin{aligned} x^T A x &= x^T Q \Lambda Q^T x \\ &= z^T \Lambda z \\ &= \sum_{i=1}^n \lambda_i z_i^2. \end{aligned}$$

Therefore,

$$\boxed{A \succeq 0 \iff \lambda_i \geq 0 \text{ for every } i,}$$

and

$$\boxed{A \succ 0 \iff \lambda_i > 0 \text{ for every } i.}$$

This representation is also the starting point for studying distributions of random quadratic forms.

26. Generalized Inverses: A Preview

In the full-rank model, $X^T X$ is invertible and

$$\hat{\beta} = (X^T X)^{-1} X^T Y.$$

When $\text{rank}(X) < p$, $X^T X$ is singular, so an ordinary inverse does not exist.

Definition 26.1 (Generalized inverse). A matrix B is a generalized inverse of A if

$$\boxed{ABA = A.}$$

A generalized inverse is often denoted A^- .

Unlike an ordinary inverse, a generalized inverse need not be unique. Nevertheless, important quantities defined using generalized inverses can be unique.

For example, if B is any generalized inverse of $X^T X$, then

$$XBX^T$$

is the orthogonal projection onto $\mathcal{C}(X)$.

Thus even when β is not uniquely identified, the fitted mean vector may still be unique.

27. Worked Regression Example

Let

$$X = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

The columns are

$$x_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad x_2 = \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}.$$

They are linearly independent, so

$$\text{rank}(X) = 2.$$

Hence $X^T X$ is invertible. We have

$$X^T X = \begin{pmatrix} 3 & 3 \\ 3 & 5 \end{pmatrix}, \quad (X^T X)^{-1} = \frac{1}{6} \begin{pmatrix} 5 & -3 \\ -3 & 3 \end{pmatrix}.$$

Therefore

$$H = X(X^T X)^{-1} X^T$$

projects any vector $Y \in \mathbb{R}^3$ onto the two-dimensional plane $\mathcal{C}(X) \subset \mathbb{R}^3$.

The fitted values and residuals are

$$\hat{Y} = HY, \quad e = (I - H)Y,$$

with

$$\hat{Y} \in \mathcal{C}(X), \quad e \in \mathcal{C}(X)^\perp, \quad \hat{Y}^T e = 0.$$

This small example contains the essential geometry of ordinary least squares.

28. Worked Rank-Deficient Example

Consider

$$X = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 2 \\ 1 & 2 & 3 \end{pmatrix}.$$

The third column satisfies

$$x_3 = x_1 + x_2,$$

so the columns are linearly dependent and

$$\text{rank}(X) = 2 < 3.$$

The nonzero vector

$$a = \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$$

satisfies

$$Xa = 0.$$

Hence for every scalar c ,

$$X(\beta + ca) = X\beta.$$

Thus the parameter vector is not uniquely determined by the mean vector.

However,

$$\mathcal{C}(X) = \text{span}(x_1, x_2)$$

is still a well-defined two-dimensional subspace, and the orthogonal projection of Y onto $\mathcal{C}(X)$ is unique.

Key idea

A central distinction in less-than-full-rank models is:

the parameterization may be nonunique while the model space is unique.

This distinction leads directly to the theory of estimable functions and generalized inverses.

29. Summary of Core Relationships

The following identities and interpretations should become familiar.

Concept	Key statement
Matrix-vector product	Ax is a linear combination of the columns of A .
Column space	$\mathcal{C}(A) = \{Ax : x \in \mathbb{R}^p\}$.
Null space	$\mathcal{N}(A) = \{x : Ax = 0\}$.
Rank	$\text{rank}(A) = \dim \mathcal{C}(A)$.
Rank-nullity	$\text{rank}(A) + \dim \mathcal{N}(A) = p$ for $A \in \mathbb{R}^{n \times p}$.
Orthogonal complement	$\mathcal{C}(A)^\perp = \mathcal{N}(A^T)$.
Gram matrix	$\mathcal{N}(X^T X) = \mathcal{N}(X)$ and $\text{rank}(X^T X) = \text{rank}(X)$.
Full-rank condition	$\text{rank}(X) = p \iff X^T X$ is invertible.
Least squares	$\hat{Y} = P_{\mathcal{C}(X)} Y$.
Hat matrix	$H = X(X^T X)^{-1} X^T$ in the full-rank case.
Projection matrix	$P^T = P$ and $P^2 = P$.
Residual projection	$e = (I - H)Y$.
Quadratic form	$x^T A x$.
Positive semidefinite	$A \succeq 0 \iff x^T A x \geq 0$ for all x .

30. Results Worth Being Able to Prove

As preparation for linear model theory, you should be comfortable proving the following results rather than merely memorizing them.

1. $\mathcal{N}(X^T X) = \mathcal{N}(X)$.
2. $\text{rank}(X^T X) = \text{rank}(X)$.
3. If X has full column rank, then $X^T X$ is positive definite.
4. $\mathcal{C}(X)^\perp = \mathcal{N}(X^T)$.
5. If $P = P^T = P^2$, then P is the orthogonal projection onto $\mathcal{C}(P)$.
6. If $P^2 = P$, every eigenvalue of P is either 0 or 1.
7. If P is symmetric and idempotent, then $\text{tr}(P) = \text{rank}(P)$.

31. Practice Problems

Problem 1: Matrix-vector multiplication

Let

$$A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \\ 1 & 3 \end{pmatrix}, \quad x = \begin{pmatrix} 2 \\ -1 \end{pmatrix}.$$

Compute Ax and express the result explicitly as a linear combination of the columns of A .

Problem 2: Linear independence

Determine whether

$$x_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \quad x_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \quad x_3 = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$$

are linearly independent.

Problem 3: Rank and null space

For

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{pmatrix},$$

find $\text{rank}(A)$ and describe $\mathcal{N}(A)$. Verify the rank-nullity theorem.

Problem 4: A fundamental null-space identity

Show that

$$\mathcal{N}(X^T X) = \mathcal{N}(X).$$

Hint: use

$$a^T X^T X a = \|Xa\|^2.$$

Problem 5: Complementary projection

Suppose P satisfies

$$P^T = P, \quad P^2 = P.$$

Show that $I - P$ is also symmetric and idempotent.

Problem 6: Orthogonality of complementary projections

Show that

$$P(I - P) = 0.$$

Interpret this result geometrically.

Problem 7: Eigenvalues of an idempotent matrix

Suppose $P^2 = P$. Show that every eigenvalue of P is either 0 or 1.

Problem 8: The hat matrix acts as the identity on the model space

Let X have full column rank and define

$$H = X(X^T X)^{-1} X^T.$$

Show that

$$HX = X.$$

Interpret this result geometrically.

Problem 9: Residual orthogonality

For the same H , show that

$$X^T(I - H) = 0.$$

Explain why this implies that least-squares residuals are orthogonal to every column of X .

Problem 10: Least-squares minimization

Suppose

$$Y = X\beta + \varepsilon$$

and X has full column rank. Show that

$$\hat{\beta} = (X^T X)^{-1} X^T Y$$

minimizes

$$(Y - X\beta)^T(Y - X\beta).$$

You may use either differentiation or the geometry of orthogonal projection.

32. Looking Ahead

The concepts introduced here lead directly into the theory developed in STAT 9120. We will use

- **column spaces** to describe the set of possible mean vectors;
- **rank and null spaces** to understand identifiability and estimability;
- **orthogonal projections** to derive least-squares estimators and ANOVA decompositions;
- **eigenvalues and spectral decompositions** to study symmetric matrices and quadratic forms; and
- **generalized inverses** to extend the theory to rank-deficient models.

The central geometric picture is

$$Y = P_{\mathcal{C}(X)}Y + (I - P_{\mathcal{C}(X)})Y,$$

or equivalently,

$$Y = \hat{Y} + e, \quad \hat{Y} \perp e.$$

Much of linear model theory can be understood as increasingly sophisticated consequences of this decomposition.